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Mechanical Engineering

PORTFOLIO

2024 – 2026

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PROFILE



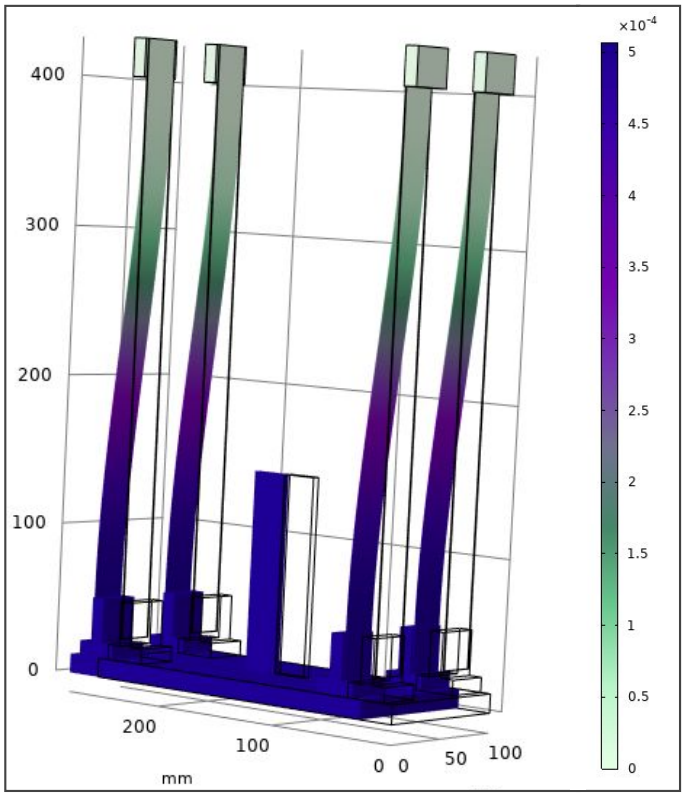
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About

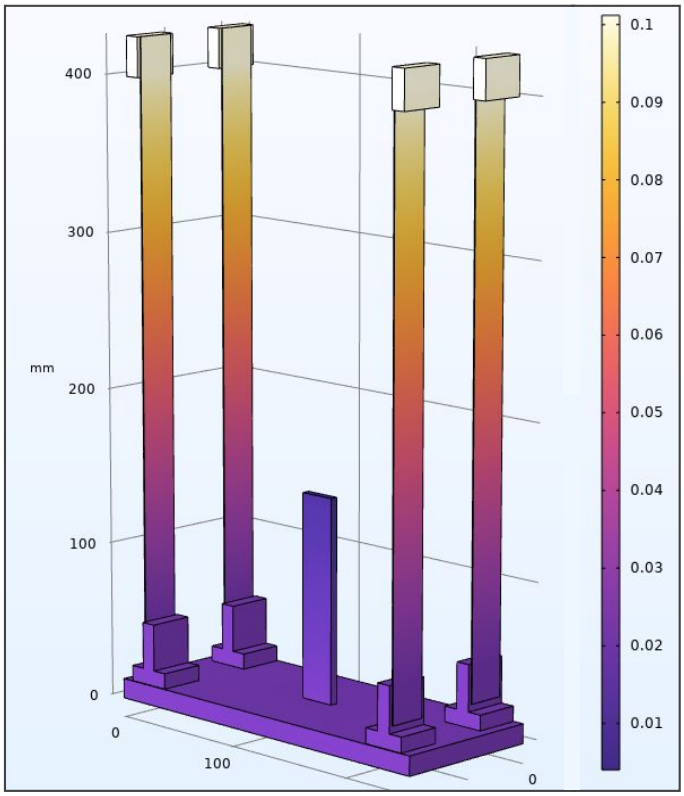
This portfolio highlights some of the engineering projects I've worked on through undergraduate research, mechanical design work, and competition robotics. As a mechanical engineering student, I value opportunities to apply technical concepts to engineering challenges involving simulation, analysis, design iteration, and real-world implementation. The projects included here span cryogenic vibration isolation research, COMSOL mechanical and thermal modeling, CAD development, and subsystem design, reflecting my continued growth in engineering problem solving, manufacturable design, and technical communication. Feel free to contact me with any questions about these projects through email.

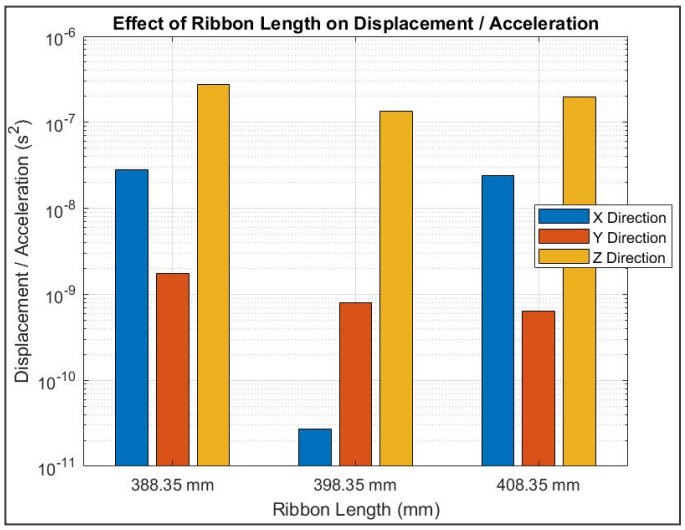
Cryogenic Vibration Isolation System (Vib-Iso) **Simulation Studies**

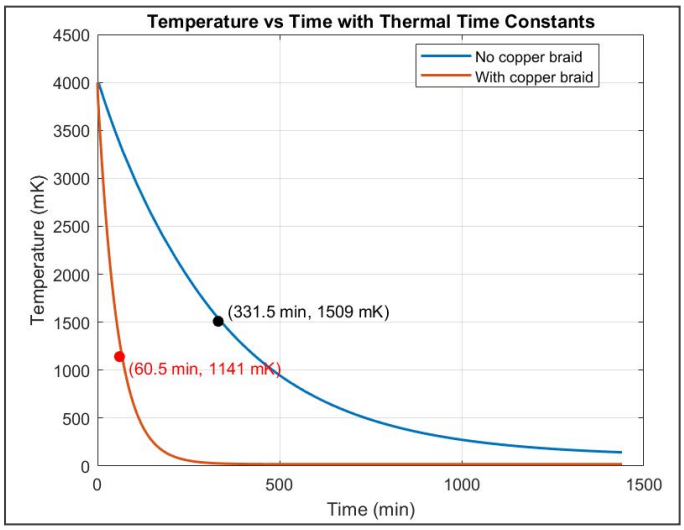
Frequency-Domain Displacement (mm × 10^{-4})



Volume Temperature Distribution (K)







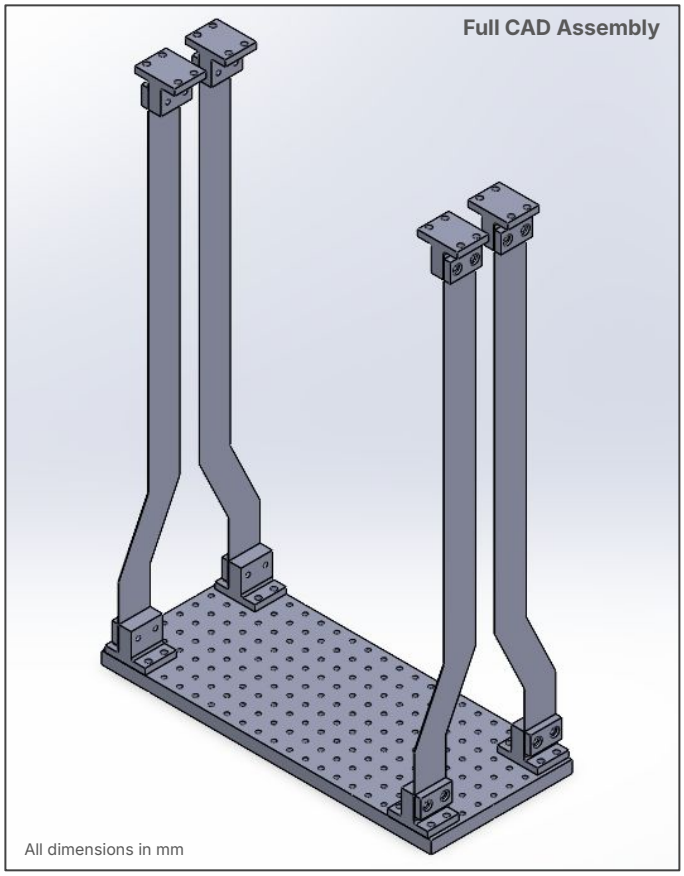
Task: Designed a vibration isolation structure for a dilution refrigerator where both mechanical stability and thermal conduction are critical. Small changes significantly impact resonance behavior and cooling performance.

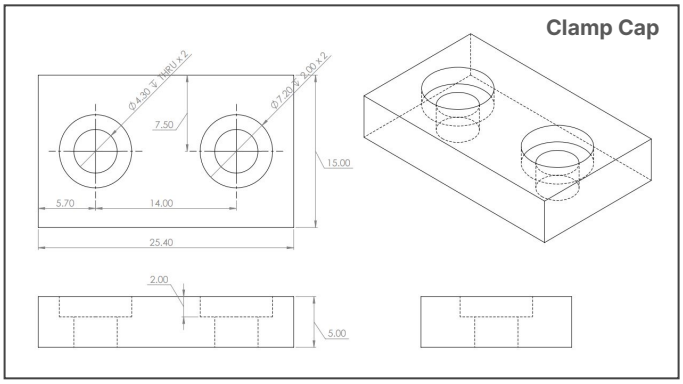
Approach & Key Insights

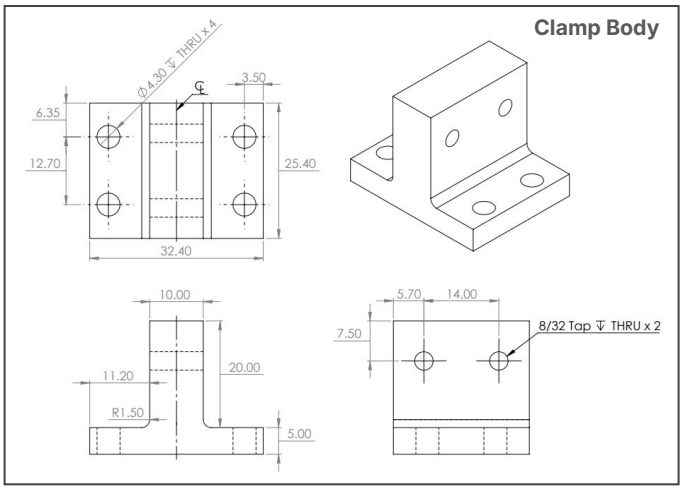
- Performed frequency-domain simulations under harmonic excitation
- Varied ribbon lengths (±10 mm) to evaluate sensitivity
- ~398 mm ribbon length minimizes displacement
- Ran stationary and time-dependent thermal studies
- Compared designs with/without braid anchoring
- Copper braid significantly improves thermal conduction and cooldown

Impact: Enabled transition from simulation to fabrication-ready Vib-Iso system for cryogenic quantum experiments

Engineering **Design & Drawings** of Vibration Isolation System









Design Process:

- Converted simplified simulation model into detailed CAD
- Selected OFHC copper for high thermal conductivity
- New functional features: mounting holes, clamp geometry, etc.
- Designed curved ribbon to preserve optical line-of-sight

Manufacturability:

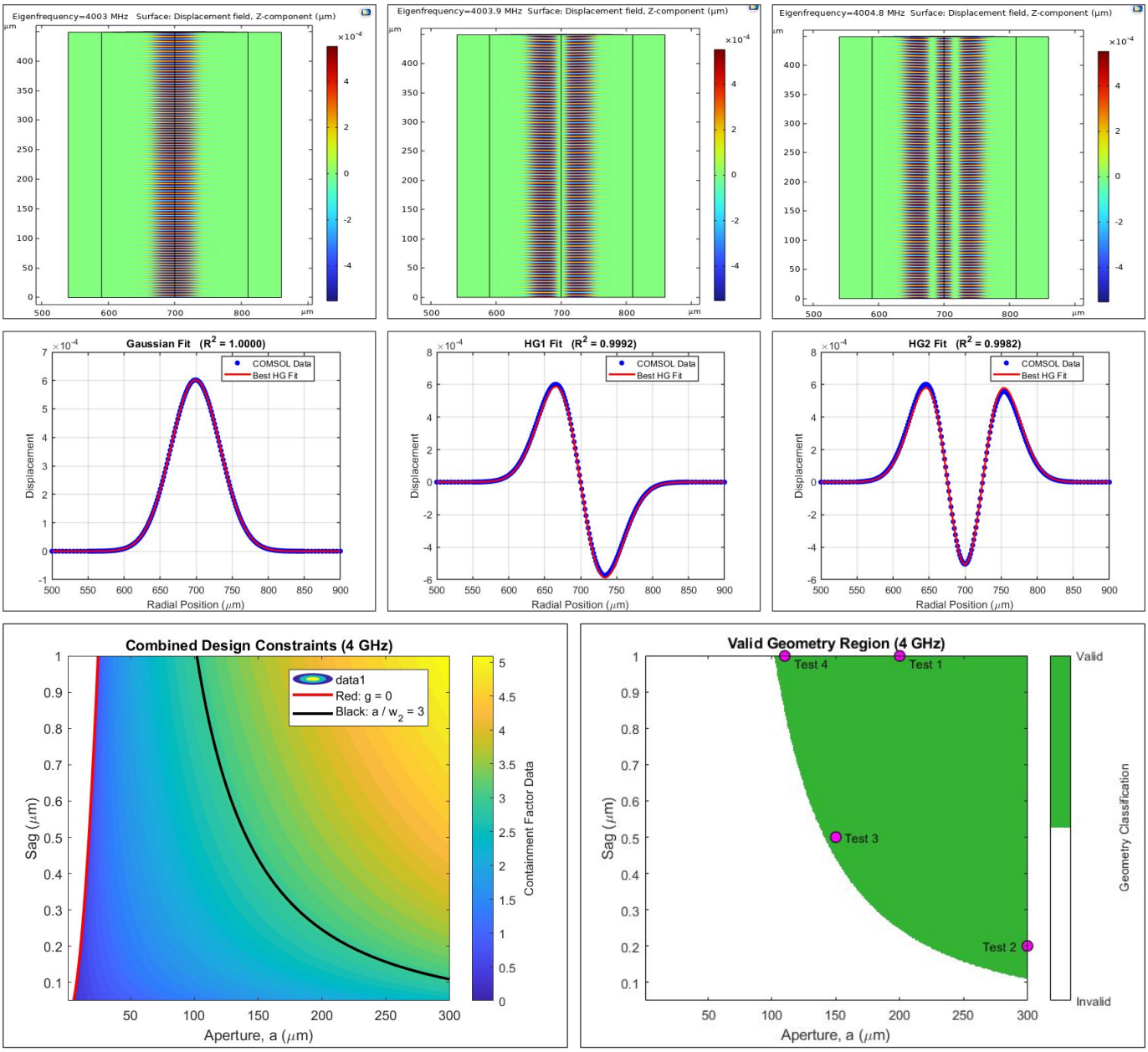
- Produced engineering drawings with tolerances and specs
- Incorporated threaded features (8-32, M3) for assembly
- Used section views and hidden lines to define internal geometry
- Optimized layouts to reduce ambiguity

Challenges & Solutions:

- First drawing set: clear dimensioning and layout practices
- Resolved ambiguity in internal features using section views
- Reduced drawing clutter through iterative refinement
- Balanced design intent with fabrication constraints

Outcome: Completed engineering drawing set for all components; parts successfully manufactured and delivered

HBAR Device Characterization & Geometry Optimization



Objective: Develop an automated COMSOL–MATLAB workflow to identify High Overtone Bulk Acoustic Resonator (HBAR) modes and optimize substrate geometry for future quantum sensing devices.

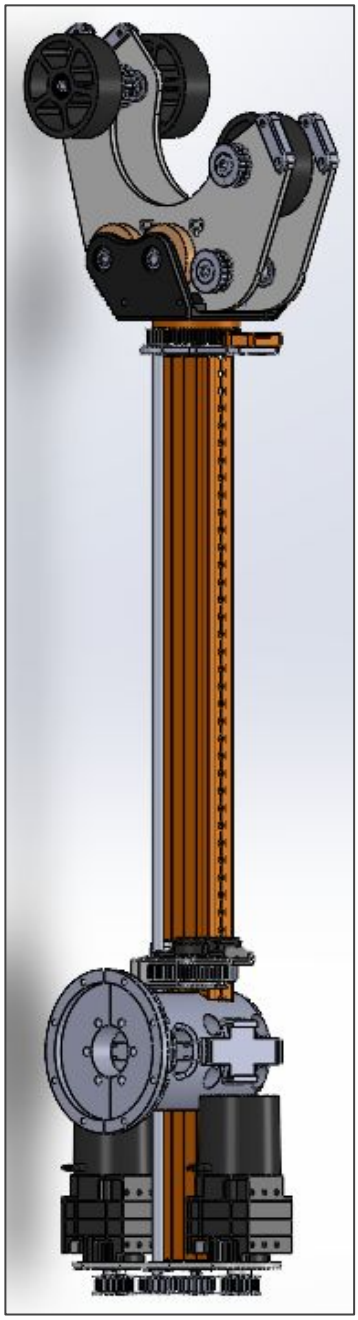
Mode Characterization:

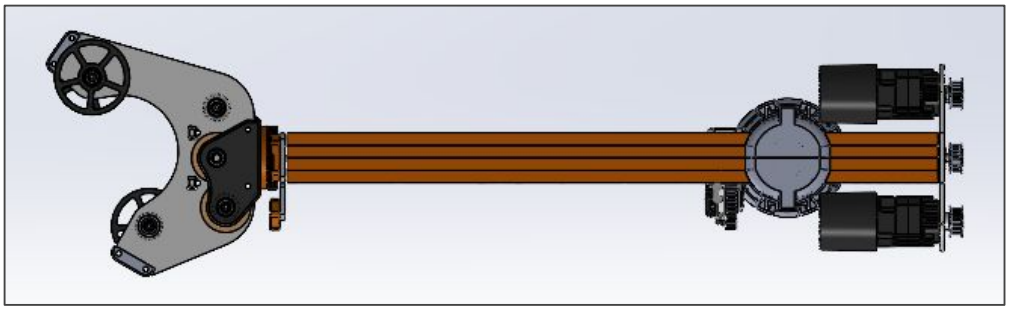
- Calculated theoretical HBAR resonance frequencies and performed COMSOL eigenfrequency searches around each predicted mode.
- Extracted displacement profiles and fit them to Hermite-Gaussian distributions using nonlinear least-squares optimization in MATLAB.
- Classified candidates using R^2 goodness-of-fit metrics and compiled the results for later optical studies.

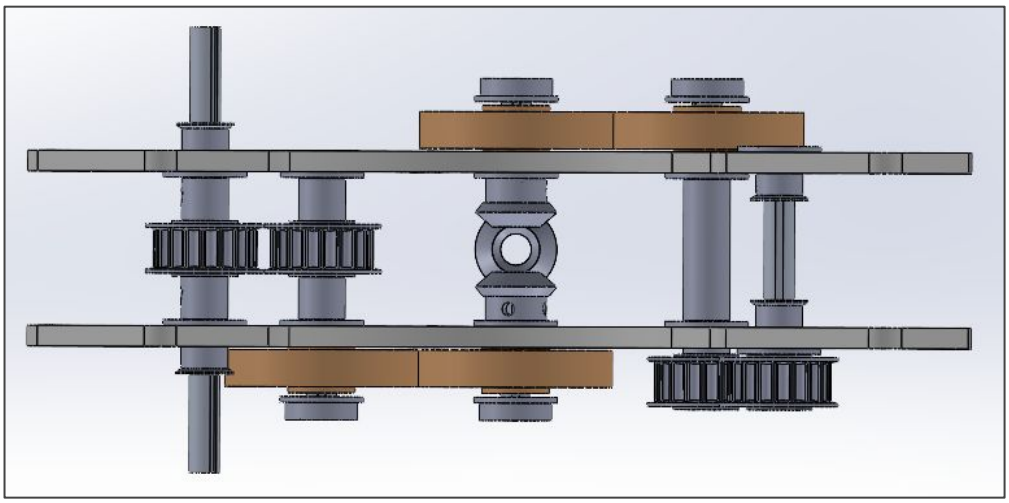
Geometry Optimization:

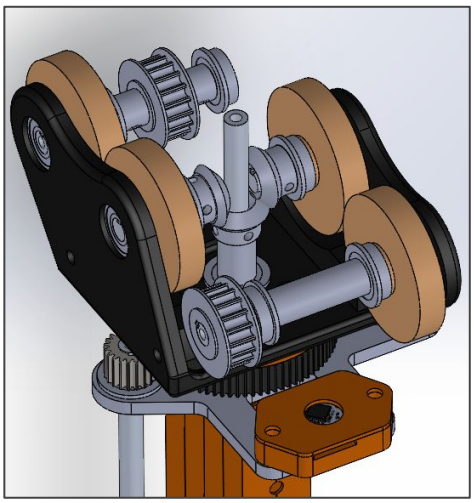
- Developed analytical MATLAB tools to evaluate how dome aperture and sag influence resonator stability, beam confinement, and operating frequency.
- Generated design-space maps to identify feasible geometries satisfying stability and confinement criteria.
- Validated optimized geometries through COMSOL simulations, confirming well-confined HBAR modes with strong agreement to Hermite-Gaussian theory.

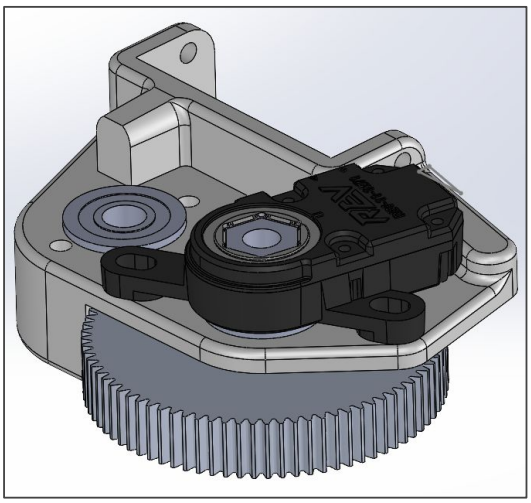
COMETS 2025 “Tempel” Differential **End Effector** Subsystem











Objective: Design an end effector capable of complex game piece manipulation through independent wrist rotation and flywheel intake for FIRST Robotics competitions.

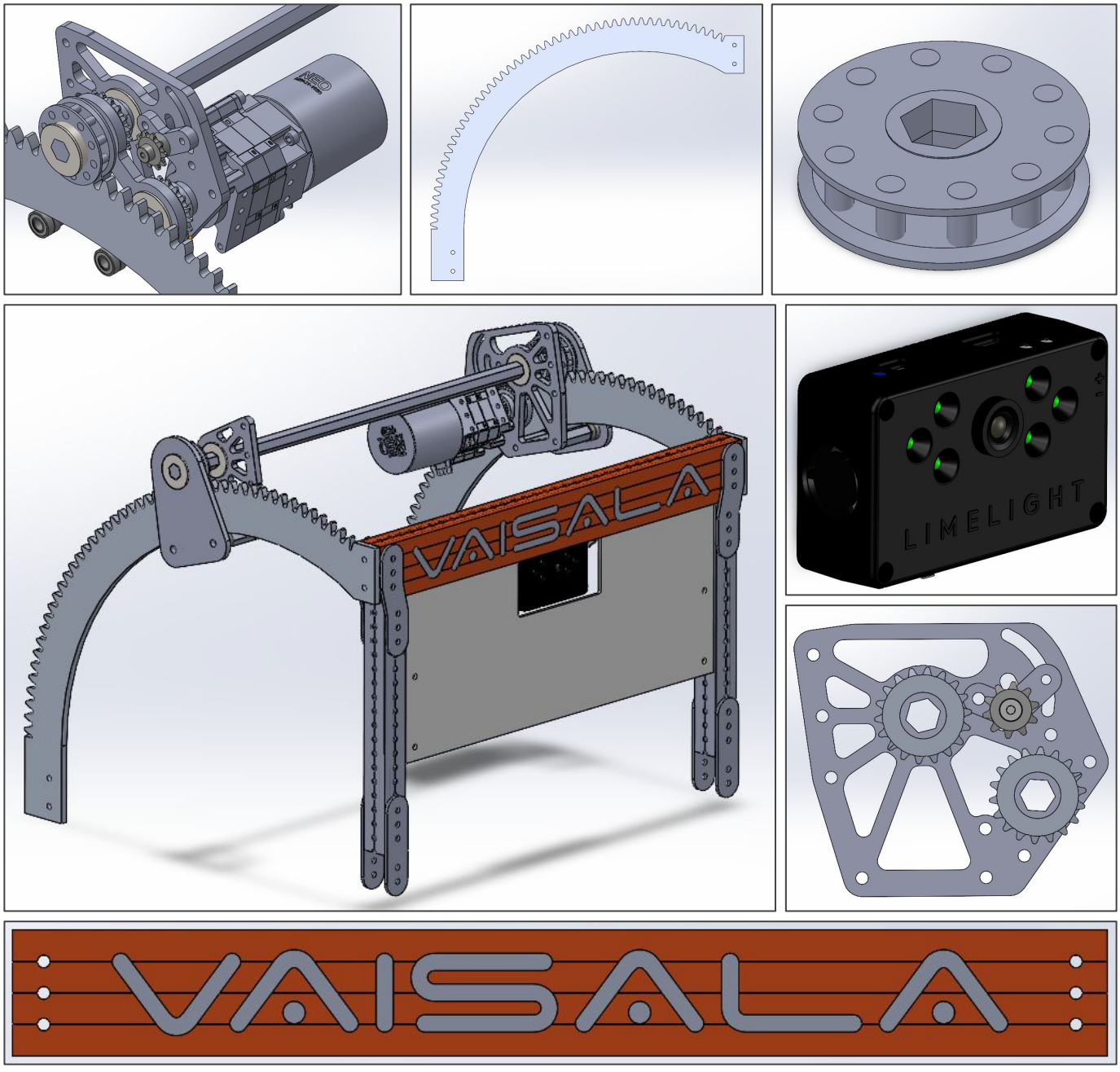
Key Components:

- Differential bevel gearbox, dual shafts for independent wrist and intake control
- Pulley-driven power transmission
- Modular assemblies for quick repair and integration
- Neo motors positioned opposite the wrist assembly for improved weight distribution

Challenges & Iteration:

- Intake geometry underwent multiple redesigns to optimize contact with game pieces
- Balanced rigidity, packaging, and rotational freedom
- Improved grip stability by transitioning from using four intake wheels to three
- Packaging required careful pulley and shaft routing

COMETS 2024 “Vaisala” Dynamic **Support Pivot** Subsystem



Objective: Design a support pivot for adjusting the launch angle of the robot’s shooting subsystem while maintaining drivetrain reliability, stiffness, and impact resistance during competition operation.

Key Components:

- Adjustable tensioning sprocket to maintain consistent tension across chain drivetrain
- Cross-shaft connected to dual pinion gears
- Large 0.25″ polycarbonate gear arcs – high durability through repeated impact loads
- Limelight vision camera for automated targeting and robot odometry

Challenges & Iteration:

- Coordinating with other robot subsystems introduced packaging and alignment constraints
- Reduced mass of structural aluminum plates through lightweighting patterns
- Manufacturing large gear arcs required careful CNC workholding and multi-pass machining
- Balanced structural rigidity with manufacturability